MOSAIC: MODAL SUPRA-ADJACENCY LEARNING FOR MULTILAYER GRAPHS WITH INTER-LAYER COUPLING

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ABSTRACT

Multiview data exhibit relational structure both within and across views. Existing multilayer graph learning methods infer the structure of individual layers but do not model how the layers relate, while product and multiplex graph learning methods capture inter-layer coupling only through a single global factor. We introduce MOSAIC, a framework that learns the full multilayer network, including the inter-layer relationships, from observed data. MOSAIC represents both intra- and inter-layer connectivity as combinations of a small set of shared patterns, or modes, capturing richer interactions compared to single global coupling. multilayer graph learning is then formulated as a regularized graph total variation minimization that jointly recovers the complete multilayer graph, the view-specific graphs, and a graph over the views. Experiments on synthetic and real multilayer networks show that MOSAIC recovers an interpretable modal organization of intra- and inter-layer structure.

***Index Terms*—** multilayer graph learning, Supra-adjacency matrix, inter-layer coupling, modal decomposition

1. INTRODUCTION

Many complex systems, e.g., social, biological, transportation, and communication systems [1], involve several types of interactions among the same entities, resulting in *multilayer* networks that preserve these distinct relationships rather than collapsing them into a single graph [2]. Such networks are represented algebraically by tensors or supra-adjacency matrices that capture both intra- and inter-layer connections [3].

Graph learning infers an unknown topology from observed signals under structural assumptions [4]. For example, smoothness-based methods recover a sparse graph on which strongly connected nodes take on similar values [5, 6]. However, these methods assume a single underlying graph with identically distributed signals, whereas many applications involve heterogeneous data that require learning multiple related graphs, i.e., multilayer graphs.

Joint graph learning methods address this limitation by inferring multiple related graphs, exploiting their shared structure through either pairwise [7, 8] or consensus-based [9, 10] similarity. These methods implicitly couple the layers through shared structure, without learning the node-to-node connections across layers.

Recently, product and multiplex graph learning approaches have imposed inter-layer connectivity through global structure. Cartesian product graph learning estimates factor graphs under a Kronecker-sum topology. When the factors are physical nodes and layers, a single physical graph is shared across all layers and inter-layer connectivity is restricted to couplings between replicas of the same node [11, 12]. Multiplex graph learning relaxes the homogeneous layer assumption and learns distinct intra-layer graphs together with a global layer coupling graph, inducing inter-layer connections through compositions of these factors [13]. In both cases, inter-layer connectivity is governed by global coupling factors rather than learned per node pair.

Relation to prior work. We introduce Modal Supra-Adjacency learning with Inter-layer Coupling (MOSAIC), a framework for learning the structured supra-adjacency matrix from multiview signals. Unlike product and multiplex formulations, where inter-layer structure is limited to a shared global coupling, the central premise of MOSAIC is that intra- and inter-layer relations are different manifestations of shared modal connectivity patterns, where each mode is associated with a layer participation profile and a modal pattern graph. The first determines how the mode is expressed across layers, and the latter determines which edges are active in that mode and their strengths. Combining these shared modes reconstructs the connectivity of each node pair across all layer pairs, which jointly yields the intra- and inter-layer adjacencies and an induced graph over layers.

2. BACKGROUND AND NOTATION

2.1. Graphs and multilayer Representation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ be a weighted, undirected simple graph with n nodes and symmetric adjacency matrix, $\mathbf{A} \in \mathbb{R}_{+}^{n \times n}$ with $\mathrm{diag}(\mathbf{A}) = \mathbf{0}$. The corresponding degree matrix is $\mathbf{D} := \mathrm{diag}(\mathbf{A}\mathbf{1}_n)$ and the combinatorial Laplacian is $\mathbf{L}(\mathbf{A}) := \mathbf{D} - \mathbf{A}$.

For a K-layer graph with physical-node set $\mathcal{V} = \{1, \ldots, n\}$,

define the node–layer set $\mathcal{V}_{\text{sup}} := \{(i,k) : i \in \mathcal{V}, k \in \{1,\dots,K\}\}$. Following standard multilayer notation [2, 3], the topology of the corresponding multilayer graph is encoded by a symmetric, hollow supra-adjacency matrix $\boldsymbol{\mathcal{A}} \in \mathbb{R}_+^{nK\times nK}$ with $\mathbf{L}_{\text{sup}} := \mathbf{L}(\boldsymbol{\mathcal{A}})$. Under layer-major ordering, $\boldsymbol{\mathcal{A}} = [\mathbf{A}^{(k\ell)}]_{k,\ell=1}^K$ with $\mathbf{A}^{(\ell k)} = (\mathbf{A}^{(k\ell)})^\top$. The diagonal blocks $\mathbf{A}^{(k)} := \mathbf{A}^{(kk)}$ are *intra-layer* adjacencies, while $\mathbf{C}^{(k\ell)} := \mathbf{A}^{(k\ell)}$, $k \neq \ell$, are *inter-layer* adjacencies. Within $\mathbf{C}^{(k\ell)}$, diagonal entries correspond to *diagonal coupling* ($i = j$) and off-diagonal entries to *non-diagonal coupling* ($i \neq j$) of nodes across layers. For a fixed pair of distinct physical nodes (i,j), define $\mathbf{P}_{ij} \in \mathbb{R}_+^{K\times K}$ as $(\mathbf{P}_{ij})_{k\ell} := \mathbf{A}_{ij}^{(k\ell)}$. Thus, $\mathbf{C}^{(k\ell)}$ encodes pairwise node relationships for a fixed pair of layers, whereas $\mathbf{P}_{ij}$ encodes the relationships for a fixed pair of nodes across layers.

2.2. Signal Smoothness

Let $\mathbf{X}^{(k)} = [\mathbf{x}_1^{(k)},\dots,\mathbf{x}_n^{(k)}]^\top \in \mathbb{R}^{n\times p}$ be the graph signals for view k and $\mathbf{X}_{\text{sup}} = [(\mathbf{X}^{(1)})^\top,\dots,(\mathbf{X}^{(K)})^\top]^\top \in \mathbb{R}^{nK\times p}$ be the multiview graph signals concatenated across the K views. A commonly used assumption in graph signal processing is the smoothness of the graph signals over the underlying graph [5]. The multilayer Dirichlet energy is defined as $\mathcal{S}(\boldsymbol{\mathcal{A}}; \mathbf{X}_{\text{sup}}) = \frac{1}{p}\operatorname{tr}\big(\mathbf{X}_{\text{sup}}^\top \mathbf{L}_{\text{sup}} \mathbf{X}_{\text{sup}}\big)$, which can be written as $\frac{1}{2p}\sum_{k,\ell=1}^K \sum_{i,j=1}^n \boldsymbol{\mathcal{A}}_{(i,k),(j,\ell)} \left\|\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(\ell)}\right\|_2^2$. In this formulation, terms with $k = \ell$ and $k \neq \ell$ quantify intra- and inter-layer smoothness, respectively.

3. PROPOSED MOSAIC FRAMEWORK

3.1. Modal Supra-Adjacency Model

We assume the connectivity matrix of each physical-node pair, (i,j) across layers, $\mathbf{P}_{ij}$, is symmetric ($\mathbf{P}_{ij} = \mathbf{P}_{ij}^\top$) and admits a shared low-rank modal representation of rank r:

$$\mathbf{P}_{ij} = \mathbf{B}\,\operatorname{diag}(\mathbf{a}_{ij})\mathbf{B}^\top = \sum_{m=1}^r a_{ij,m}\mathbf{b}_m\mathbf{b}_m^\top, \qquad (1)$$

where the columns of $\mathbf{B} := [\mathbf{b}_1,\dots,\mathbf{b}_r]$ are the nonnegative layer participation profiles, $\mathbf{a}_{ij} := [a_{ij,1},\dots,a_{ij,r}]^\top \in \mathbb{R}_+^r$ is the pair-specific modal strength vector where $a_{ij,m}$ specifies the strength of mode m for physical-node pair (i,j). Collecting the coefficients, $a_{ij,m}$, yields a symmetric, hollow modal pattern graph $\mathbf{A}_m$ with $(\mathbf{A}_m)_{ij} = (\mathbf{A}_m)_{ji} = a_{ij,m}$ for $i < j$. The layer participation profile $\mathbf{b}_m \in \mathbb{R}_+^K$ describes the participation of each layer in mode m. To avoid scale ambiguity, we impose $\mathbf{b}_m \in \Delta_K := \{\mathbf{b} \in \mathbb{R}_+^K : \mathbf{1}_K^\top \mathbf{b} = 1\}$.

To model diagonal couplings, let $\mathbf{c}_m := [c_{1,m},\dots,c_{n,m}]^\top \in \mathbb{R}_+^n$, $\mathbf{D}_m := \operatorname{diag}(\mathbf{c}_m)$, and $\mathbf{M}_m := \operatorname{offdiag}(\mathbf{b}_m\mathbf{b}_m^\top)$. For $\mathcal{Z} := \{\mathbf{A}_m, \mathbf{D}_m, \mathbf{b}_m\}_{m=1}^r$, the supra-adjacency matrix can be written as

$$\boldsymbol{\mathcal{A}}(\mathcal{Z}) = \sum_{m=1}^r (\mathbf{b}_m\mathbf{b}_m^\top) \otimes \mathbf{A}_m + \sum_{m=1}^r \mathbf{M}_m \otimes \mathbf{D}_m. \qquad (2)$$

The first term corresponds to all edges between distinct node pairs, i.e., $i \neq j$: intra-layer for $k = \ell$ and inter-layer for $k \neq \ell$. The second term corresponds to the inter-layer edges for $i = j$. Consequently, the intra-layer adjacency is $\mathbf{A}^{(k)} = \sum_{m=1}^r b_{k,m}^2 \mathbf{A}_m$, while the inter-layer adjacency is $\mathbf{C}^{(k\ell)} = \sum_{m=1}^r b_{k,m}b_{\ell,m}(\mathbf{A}_m + \mathbf{D}_m)$ for $k \neq \ell$.

Aggregating over inter-layer adjacency matrices yields an induced graph over layers with adjacency matrix, $\mathbf{G}_{\text{layer}} \in \mathbb{R}_+^{K\times K}$, with diagonals equal to zero and, for $k \neq \ell$, $(\mathbf{G}_{\text{layer}})_{k\ell} := \mathbf{1}_n^\top \mathbf{C}^{(k\ell)} \mathbf{1}_n$. Hence, the same modal connectivity patterns determine the intra-layer graphs, both types of inter-layer connections, and the graph over layers.

3.2. Objective Function

Let $E = n(n-1)/2$ and enumerate the physical-node pairs as $e = (i_e, j_e)$, $i_e < j_e$. We define $\boldsymbol{\Theta} \in \mathbb{R}_+^{E\times r}$, where $\Theta_{e,m} = (\mathbf{A}_m)_{i_e j_e}$ and $\boldsymbol{\Gamma} \in \mathbb{R}_+^{n\times r}$, with $\Gamma_{i,m} := c_{i,m}$, and $\mathbf{B} = [\mathbf{b}_1,\dots,\mathbf{b}_r] \in \Delta_K^r$, with $\mathbf{D}_m = \operatorname{diag}(\boldsymbol{\Gamma}_{:,m})$. Hence, $\boldsymbol{\Theta}$, $\boldsymbol{\Gamma}$, and $\mathbf{B}$ provide a one-to-one parameterization of $\{\mathbf{A}_m, \mathbf{D}_m, \mathbf{b}_m\}$.

Using the unsigned node–edge incidence matrix $\mathbf{H} \in \{0,1\}^{n\times E}$, we define $\mathbf{Q} := \mathbf{H}\boldsymbol{\Theta}, \mathbf{U} := \mathbf{B} \odot \mathbf{B}, \mathbf{W} := \mathbf{B} \odot (\mathbf{1}_{K\times r} - \mathbf{B})$. The intra- and inter-layer degree matrices can then be written as $\mathbf{D}_{\text{intra}} = \mathbf{Q}\mathbf{U}^\top, \mathbf{D}_{\text{inter}} = (\mathbf{Q} + \boldsymbol{\Gamma})\mathbf{W}^\top$, whose (i,k)th entries are $d_{i,k}^{\text{intra}}$ and $d_{i,k}^{\text{inter}}$, respectively.

To rewrite the signal smoothness term in terms of these variables, we denote $\delta_{ij}^{k\ell} := p^{-1}\|\mathbf{x}_i^{(k)} - \mathbf{x}_j^{(\ell)}\|_2^2$ and define $(\mathbf{S}_e)_{k\ell} := \delta_{i_e j_e}^{k\ell}$. With these definitions, the objective function of MOSAIC can be expressed as:

$$\begin{aligned}
\min_{\boldsymbol{\Theta},\boldsymbol{\Gamma},\mathbf{B}} \quad & \mathcal{F}(\boldsymbol{\Theta},\boldsymbol{\Gamma},\mathbf{B}) = \langle \boldsymbol{\Theta}, \boldsymbol{\Xi}^a(\mathbf{B})\rangle + \langle \boldsymbol{\Gamma}, \boldsymbol{\Xi}^c(\mathbf{B})\rangle \\
& + \alpha_a \mathbf{1}_E^\top \boldsymbol{\Theta}\mathbf{1}_r + \frac{\beta_a}{2}\|\boldsymbol{\Theta}\|_F^2 + \alpha_c \mathbf{1}_n^\top \boldsymbol{\Gamma}\mathbf{1}_r + \frac{\beta_c}{2}\|\boldsymbol{\Gamma}\|_F^2 \\
& - \tau_N \sum_{i=1}^n \sum_{k=1}^K \log(d_{i,k}^{\text{intra}} + \varepsilon_N) \\
& - \tau_I \sum_{i=1}^n \sum_{k=1}^K \log(d_{i,k}^{\text{inter}} + \varepsilon_I) + \lambda_B \sum_{m<q} (\mathbf{b}_m^\top \mathbf{b}_q)^2 \\
\text{s.t.} \quad & \boldsymbol{\Theta} \geq \mathbf{0}, \quad \boldsymbol{\Gamma} \geq \mathbf{0}, \quad \mathbf{b}_m \in \Delta_K, \quad m = 1,\dots,r.
\end{aligned} \qquad (3)$$

where $[\boldsymbol{\Xi}^a(\mathbf{B})]_{e,m} := \mathbf{b}_m^\top \mathbf{S}_e \mathbf{b}_m$, $[\boldsymbol{\Xi}^c(\mathbf{B})]_{i,m} := \frac{\eta_c}{2}\mathbf{b}_m^\top \mathbf{R}_i \mathbf{b}_m$ and $(\mathbf{R}_i)_{k\ell} := \delta_{ii}^{k\ell}\mathbb{I}\{k \neq \ell\}$ with $\mathbb{I}\{\cdot\}$ being the indicator function. The first two terms correspond to the smoothness with respect to the supra-adjacency matrix, with η_c controlling the relative contribution of diagonal inter-layer coupling. When $\eta_c = 1$, this reduces to the supra-adjacency Dirichlet energy. α_a, α_c terms promote sparsity, the quadratic terms

control graph scale, the log-degree terms discourage isolated intra- and inter-layer components, and the λ_B term ensures that the modal layer profiles do not overlap. We assume $\beta_a, \beta_c, \varepsilon_N, \varepsilon_I > 0$, with the remaining regularization parameters nonnegative.

3.3. Optimization

MOSAIC solves (3) by projected block coordinate descent over $\mathcal{C}_\Theta := \mathbb{R}_+^{E\times r}, \mathcal{C}_\Gamma := \mathbb{R}_+^{n\times r}, \mathcal{C}_B := \Delta_K^r$, with $\mathcal{C} := \mathcal{C}_\Theta \times \mathcal{C}_\Gamma \times \mathcal{C}_B$ and at iteration t, $\boldsymbol{\mathcal{Z}}^t := (\boldsymbol{\Theta}^t, \boldsymbol{\Gamma}^t, \mathbf{B}^t)$. Using the most recent updates from the preceding block, one projected gradient step is applied to each variable in the following order:

$$\boldsymbol{\Theta}^{t+1} = \mathrm{Proj}_{\mathcal{C}_\Theta}\left(\boldsymbol{\Theta}^t - (L_\Theta^t)^{-1}\nabla_\Theta \mathcal{F}(\boldsymbol{\Theta}^t, \boldsymbol{\Gamma}^t, \mathbf{B}^t)\right), \quad (4)$$

$$\boldsymbol{\Gamma}^{t+1} = \mathrm{Proj}_{\mathcal{C}_\Gamma}\left(\boldsymbol{\Gamma}^t - (L_\Gamma^t)^{-1}\nabla_\Gamma \mathcal{F}(\boldsymbol{\Theta}^{t+1}, \boldsymbol{\Gamma}^t, \mathbf{B}^t)\right), \quad (5)$$

$$\mathbf{b}_m^{t+1} = \mathrm{Proj}_{\Delta_K}\left(\mathbf{b}_m^t - (L_B^t)^{-1}\mathbf{g}_m^t\right), \quad m = 1, \ldots, r, \quad (6)$$

where $\mathbf{g}_m^t := \nabla_{\mathbf{b}_m}\mathcal{F}(\boldsymbol{\Theta}^{t+1}, \boldsymbol{\Gamma}^{t+1}, \mathbf{B}^t)$. The simplex projection in (6) is computed as in [14]. The block-specific step sizes $(L_\Theta^t)^{-1}, (L_\Gamma^t)^{-1}, (L_B^t)^{-1}$ are selected by backtracking. For block index $j \in \{\Theta, \Gamma, B\}$ of $\boldsymbol{\mathcal{Z}}$, let $\widehat{\mathcal{F}}_j^t$ denote $\mathcal{F}$ with the other blocks fixed at their most recent values, and define the projected candidate $\widetilde{\boldsymbol{\mathcal{Z}}}_j^t = \mathrm{Proj}_{\mathcal{C}_j}\left(\boldsymbol{\mathcal{Z}}_j^t - (L_j^t)^{-1}\nabla_j \widehat{\mathcal{F}}_j^t(\boldsymbol{\mathcal{Z}}_j^t)\right)$. Starting from $L_j^t > \sigma_j > 0$, the candidate $\widetilde{\boldsymbol{\mathcal{Z}}}_j^t$ is accepted when the Armijo condition of [15] is satisfied; otherwise $L_j^t \leftarrow \rho_{\mathrm{bt}} L_j^t$, $\rho_{\mathrm{bt}} > 1$, and the candidate is recomputed. Iterations terminate when the projected gradient residuals $r_j^t := L_j^t \left\| \boldsymbol{\mathcal{Z}}_j^t - \widetilde{\boldsymbol{\mathcal{Z}}}_j^t \right\|_F$ and the relative objective change satisfy $\max_{j\in\{\Theta,\Gamma,B\}} r_j^t \le \varepsilon_{\mathrm{stat}}$ and $r_{\mathrm{obj}}^t \le \varepsilon_{\mathrm{obj}}$. The implementation is available online[1].

Convergence: For $\beta_a, \beta_c, \varepsilon_N, \varepsilon_I > 0$, F is bounded below with locally Lipschitz gradient. $\Phi = F + \iota_{\mathcal{C}}$, where $\iota_{\mathcal{C}}$ is the indicator function, satisfies the Kurdyka–Lojasiewicz property [16]. The updates then meet the assumptions of [15, Thm. 2], giving $\boldsymbol{\mathcal{Z}}^t \to \boldsymbol{\mathcal{Z}}^\star$ with $0 \in \nabla\mathcal{F}(\boldsymbol{\mathcal{Z}}^\star) + N_{\mathcal{C}}(\boldsymbol{\mathcal{Z}}^\star)$.

4. EXPERIMENTS

4.1. Synthetic Data

We generate a multilayer network with $n = 16$ nodes, $K = 4$ layers, and $r = 2$ modes. The binary modal graphs $\mathbf{A}_1$ and $\mathbf{A}_2$ consist of two communities spanning nodes 1–8 and 9–16, respectively, with within-community edge probability $q_{\mathrm{in}} = 0.19$ and between-community edge probability $q_{\mathrm{out}} = 0.01$. Diagonal couplings exist for nodes $1-4$ for mode 1 and $13-16$ for mode 2. The normalized layer participation profiles of the modes are proportional to $(1, 1-\rho, \rho, 0)$ and $(0, \rho, 1-\rho, 1)$, with default mode-mixing parameter $\rho_0 = 0.05$. Increasing ρ increases the mixing of the modes across layers which also results in stronger non-diagonal inter-layer coupling. From the resulting supra-adjacency matrix in (2), we generate $p = 5000$ signals as $\mathbf{X}_{\mathrm{sup}} = \exp(-\tau_h \mathbf{L}_{\mathrm{sup}})\mathbf{Z}$, where $\mathbf{Z} \in \mathbb{R}^{nK\times p}$ has i.i.d. Gaussian entries. We compare MOSAIC with product graph learning (PGL) [11] and multiplex graph learning (MXGL) [13].For each setting, we use ten validation and ten independent test realizations, selecting hyperparameters by grid search to maximize the mean validation F1 across intra- and inter-layer coupling components.

We evaluate the methods under four settings: (a) varying the number of graph signals p; (b) varying the average modal density through edge additions; (c) varying ρ with fixed $\mathbf{A}_m$; and (d) varying the signal-to-noise ratio and the noise heteroscedasticity parameter κ, which interpolates between i.i.d. and view-dependent noise variances. In Fig. 1(a), MOSAIC's advantage over the baselines widens as p grows, since non-diagonal inter-layer coupling is parameterized per node pair, so it needs the most samples to recover. PGL and MXGL cannot represent this coupling at all. Additional signals therefore benefit MOSAIC precisely where the baselines are structurally limited. Fig. 1(b) shows that increasing modal density affects all methods, most notably MOSAIC and PGL, while MOSAIC still performs the best. MOSAIC estimates more edge-specific connections as density increases, and PGL cannot represent off-diagonal cross-layer coupling, while MXGL stays more stable due to its strongly shared global coupling structure. In Fig. 1(c), MOSAIC maintains high recovery accuracy as ρ increases, while the performance gap widens, particularly for PGL as non-diagonal inter-layer coupling becomes stronger. Finally, in Fig. 1(d), MOSAIC degrades as SNR decreases under both i.i.d. and heteroscedastic noise, but view-dependent noise breaks the shared connectivity strength assumption that the trace-constrained baselines rely on, whereas MOSAIC's per-(i,k) log-degree barriers adapt to view-specific strengths. As a result, MOSAIC remains more robust as heteroscedasticity increases, outperforming PGL and MXGL even at 0 dB.

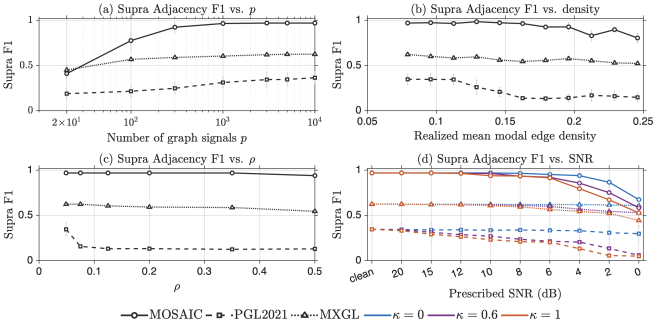


Fig. 1. Supra-adjacency F1 versus (a) number of signals p, (b) modal density, (c) mixing parameter ρ, and (d) SNR at different levels of view-wise noise heteroscedasticity κ.

[1] https://github.com/Efe6402/ICASSP2027_MOSAIC

4.2. Real-Data Experiments

4.2.1. COIL-20

We evaluate MOSAIC on COIL-20 [17], which contains 20 objects captured at 5-degree increments over a full rotation. Objects define nodes, rotation angles define layers, and the 32×32 vectorized images yield graph signals $\mathbf{X}^{(k)} \in \mathbb{R}^{20\times 1024}$. We build a six-layer network at $(\phi, \phi+60°, \ldots, \phi+300°)$, where ϕ denotes the phase offset. We determine the unknown mode count r and hyperparameters h using the train–validation phase pairs $\phi = (5°, 10°)$, $(25°, 30°)$, and $(45°, 50°)$. We average their normalized validation smoothness scores as $\bar{V}(r, h)$, define $V^{\star}(r) = \min_h \bar{V}(r, h)$ and select r as its geometric elbow. We then refit the corresponding best hyperparameter configuration at $\phi = 0°$ for testing.

MOSAIC reveals three modes for validation set, where the activations are separated by $180°$, indicating that opposite viewing angles share similar connectivity structure (Fig. 2(a)). The intra-layer graph in Fig. 2(b) captures visual similarities among different objects within the same viewing angle. The learned modal organization is also reflected in the non-diagonal inter-layer coupling in Fig. 2(c), where different objects, e.g. cars, boxes and bottles, that have natural symmetry are connected to their visually similar variants observed at $180°$ offsets. Consequently, the induced graph over layers in Fig. 2(d) groups layers separated by $180°$, consistent with the learned modal organization.

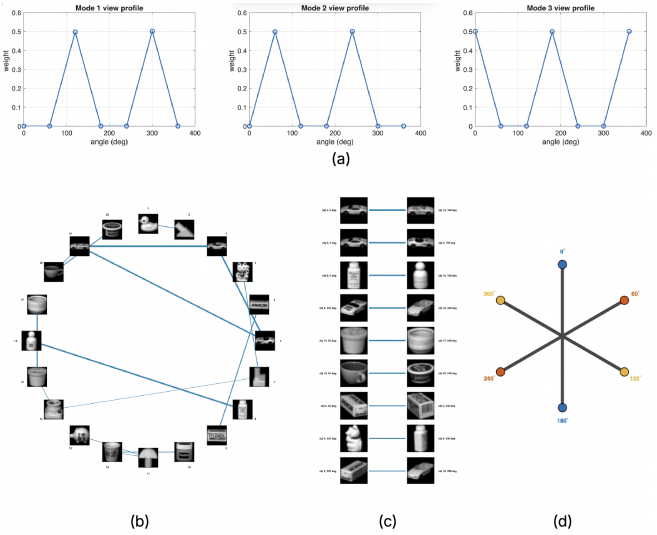


Fig. 2. MOSAIC results on COIL-20: (a) Learned modal layer participation weights $\mathbf{b}_m$ for three modes; (b) Top-15 intra-layer connections at the $0°$ layer; (c) Top-9 non-diagonal connections; (d) Clustering based on the induced graph over layers.

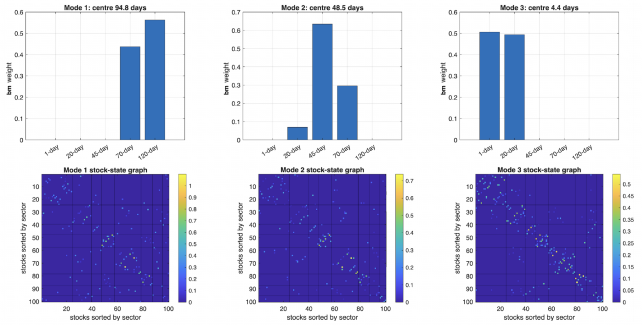


Fig. 3. Learned modal layer participation weights $\mathbf{b}_m$ and modal connectivity patterns $\mathbf{A}_m$ for S&P 500 data.

4.2.2. S&P 500

For the second dataset, we use daily S&P 500 prices [18] for $n = 100$ stocks with complete price histories. $K = 5$ layers correspond to return horizons $h_k \in \{1, 20, 45, 70, 120\}$ trading days. For horizon h_k, stock i is represented by cumulative log-returns $\log P_i(t_q) - \log P_i(t_q - h_k)$ at common endpoints $t_q = 121 + 5(q-1)$. For the final offset-0 configuration, these endpoints span June 24, 2016–December 31, 2019, giving $p = 178$ and $\mathbf{X}^{(k)} \in \mathbb{R}^{100\times 178}$. Each stock is standardized separately within each layer. For model selection, five endpoint offsets $(0, \ldots, 4)$ are formed and the hold-out selection procedure discussed for COIL-20 is used, with offset 0 reserved for the testing.

As shown in Fig. 3, the learned $\mathbf{b}_m$s separate short-, intermediate-, and long-horizon regimes, while the corresponding $\mathbf{A}_m$s evolve from predominantly sector-localized connectivity at shorter horizons ($\mathbf{A}_3$) toward more diffuse cross-sector structure at longer horizons ($\mathbf{A}_1$), consistent with the multiscale S&P 500 observations in [19].

5. CONCLUSION

We introduced MOSAIC, a modal framework for learning structured supra-adjacency matrices from multiview signals. By representing physical-node interactions as edge-specific mixtures of shared connectivity modes, MOSAIC jointly captures intra-layer structure and both diagonal and non-diagonal inter-layer coupling, while inducing a graph over layers. Synthetic experiments demonstrate accurate and robust supra-adjacency recovery under varying graph density, modal-mixing, and noise conditions, while COIL-20 and S&P 500 reveal interpretable modal and inter-layer structure. Future work will extend MOSAIC to directed multilayer graphs through asymmetric modal representations of intra- and inter-layer connectivity.

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